B06 Integrated Bridge Capstone Brief

Home Services Network | implementation-ready Bridge evidence

Scenario

Home Services Network needs one trusted analytical system for service-order economics, technician performance/history, operational events, regional targets and executive reporting. Inputs come from mixed private and cloud sources.

Your job

Act as the analyst-to-data-engineering bridge. You are not expected to deploy the Course 3 stack. Produce implementation-ready evidence that another data engineer could safely continue.

Required artifacts

Area	Required evidence
SQL	Completed SQL workspace + independent controls
Python	Completed pipeline notebook + validation/log/run evidence
Model	Facts/dimensions/bridge + keys/cardinality/date/history design
Warehouse	SCD + business/surrogate key + layers + replay/backfill reasoning
Quality	>=8 meaningful checks with severity/owner/publish consequence
Platform	Connection/access/freshness/monitoring decisions
Handoff	Professional handoff + 60-90 second defense

Minimum technical proof

1. Demonstrate one wrong-but-plausible fan-out and the control that catches it.
2. Reconcile one order-level metric to item-grain detail.
3. Use at least one window calculation and one time/LAG comparison.
4. Detect an orphan/duplicate/history-quality issue.
5. Build one reproducible multi-file Python flow with source lineage.
6. Demonstrate one critical failure that blocks trusted publish.
7. Prove one unchanged safe rerun.
8. Make one historical technician attribute decision with fact lookup rule.
9. Distinguish private/on-prem from cloud connection treatment.
10. Walk through monitoring + recovery + owner/next action.

Objective practice controls

The supplied practice control file declares 24 customers, 12 current technicians, 17 technician-change rows, 72 orders, 54 completed orders, completed base-fee total 24,150, 144 item rows, item-value total 81,600, 271 event rows with 1 orphan event, 12 target rows, 15 technician-skill rows, and 1 duplicate technician/effective-date change pair. Use independent calculations to confirm; do not trust the declaration blindly.

Failure drills

Use service_orders_bad_numeric.csv (contains BAD_AMOUNT) and technicians_duplicate.csv (duplicate T006). Record detection -> containment -> root cause evidence -> repair -> rerun/backfill -> post-recovery validation -> owner/communication.

What not to do

Do not flatten every grain into one table, hide duplication with DISTINCT, silently coerce/drop malformed records and still publish trusted output, overwrite historical truth with current attributes, invent gateways for cloud sources, or treat "the code ran" as proof.

Submission & route

Submission checklist

SQL saved; notebook cleanly rerun and saved; evidence workbook complete; source and curated row/value controls reconciled; failure/recovery evidence saved; known limitation stated; owner/next action stated; changed transfer completed.

Protected assessment integrity

This Home Services capstone is guided. The B06 Final Gate is fresh and independent. During the active Gate, lesson/review/walkthrough/AI solving stays closed. Save the first attempt before review.

Finished when

B06-L01 is Mastered only after complete evidence + changed transfer. Then begin B06-L02. Do not mark C3 ENTRY READY from this guided capstone alone.